Statistical Methods

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Outline I

1. Random Variables
2. Standard Distributions

Why Do We Need Random Variables?

Up to now, we have defined events as qualitative subsets of a sample space:

$$\mathcal{S} = \{\text{Heads}, \text{Tails}\} \quad \text{or} \quad \mathcal{S} = \{\text{Red}, \text{Blue}, \text{Green}\}$$

The Limitation of Qualitative Outcomes

- We cannot perform algebraic operations (like addition, averaging, or scaling) on raw outcomes like "Heads" or "Red".
- To build mathematical models, calculate averages, or measure variances, we must translate these outcomes into numerical values.

We need a systematic rule that assigns a real number to every single outcome in our sample space.

Formal Definition of a Random Variable

Definition

Random Variable Let $\mathcal{S}$ be the sample space of a random experiment. A **random variable** X is a real-valued function that maps every outcome $s \in \mathcal{S}$ to a unique real number $X(s) \in \mathbb{R}$.

$$X : \mathcal{S} \to \mathbb{R}$$

Crucial Conceptual Caveat

Despite its name, a random variable is **not a variable** and is **not random** in the traditional sense.

- It is a deterministic, mathematically defined **function**.
- The "randomness" comes solely from the underlying experiment which determines which outcome $s \in \mathcal{S}$ occurs.

The Range of a Random Variable

The collection of all numerical values that a random variable can actually take is called its **range** or **support**, often denoted by $\mathcal{T}$ or $X(\mathcal{S})$.

$$\mathcal{T} = \{x \in \mathbb{R} \mid X(s) = x \text{ for some } s \in \mathcal{S}\}$$

Example: Tossing Two Fair Coins

Let the sample space be:

$$\mathcal{S} = \{HH, HT, TH, TT\}$$

Let X be the random variable representing the **number of Heads**.

- $X(HH) = 2$
- $X(HT) = 1$
- $X(TH) = 1$
- $X(TT) = 0$

The support of X is the discrete set: $\mathcal{T} = \{0, 1, 2\}$.

Discrete vs. Continuous Random Variables

Depending on the nature of their support set $\mathcal{T}$, random variables are classified into two fundamental domains:

1. Discrete Random Variables

A random variable is **discrete** if its range $\mathcal{T}$ consists of a finite or countably infinite set of values (such as integers).

- Outcomes can be listed in a sequence: $\{x_1, x_2, x_3, \dots\}$.

2. Continuous Random Variables

A random variable is **continuous** if its range $\mathcal{T}$ consists of an uncountable interval or union of intervals of real numbers.

- It can assume any real-valued measurement in its range.

Discrete Example: Rolling Two Dice

Suppose two fair, six-sided dice are rolled, yielding a sample space $\mathcal{S}$ of 36 equally likely outcomes.

Let Y be the random variable representing the **sum of the two dice**:

$$Y(i,j) = i + j \quad \text{where } (i,j) \in \mathcal{S}$$

- The minimum possible value is: $Y(1,1) = 2$.
- The maximum possible value is: $Y(6,6) = 12$.
- The range is the discrete list of values:

$$\mathcal{T} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Since the range is finite, Y is a **discrete random variable**.

Continuous Example: Device Lifetime

Consider an experiment where we observe the operational lifetime (in hours) of a continuous electronic component until it experiences a structural failure.

The physical sample space consists of all non-negative real numbers:

$$\mathcal{S} = \{t \in \mathbb{R} \mid t \geq 0\}$$

Let T be the random variable representing the **lifetime of the component**:

$$T(t) = t \quad \text{for any } t \in \mathcal{S}$$

- Since the time of failure can be any fractional value (e.g., 104.532 hours), the range is the continuous interval:

$$\mathcal{T} = [0, \infty)$$

Because this range is uncountable, T is a **continuous random variable**.

Exercise: Classifying Random Variables

For each of the following scenarios, identify:

1. The random variable X of interest.
2. Whether X is **discrete** or **continuous**.
3. The mathematically defined range/support $\mathcal{T}$ of X.

Scenarios

- **Scenario A:** A fair coin is tossed repeatedly until the first "Heads" appears. We count the total number of tosses required.
- **Scenario B:** A researcher measures the precise blood pressure level (in mmHg) of a patient.
- **Scenario C:** A commuter records the exact waiting time (in minutes) for a city bus, which is guaranteed to arrive within 30 minutes.

The Induced Probability Map

A random variable $X : \mathcal{S} \to \mathbb{R}$ is a completely deterministic, algebraic function. It does not possess any "inherent" randomness.

Where does the probability come from?

The probability of X taking on a specific value is entirely **derived** from the probability of the underlying outcomes in the sample space $\mathcal{S}$ that map to that value.

If the underlying physical experiment is performed and yields an outcome $s \in \mathcal{S}$, the random variable evaluates to the number $X(s)$.

Translating Events to Real Numbers

Let x be a real number. We want to define what $P(X = x)$ means.

We look backward at the sample space $\mathcal{S}$ and gather all outcomes s that map to x. We call this subset the **pre-image** of x:

$$E_x = \{s \in \mathcal{S} \mid X(s) = x\}$$

Definition

The Probability Law of X The probability that the random variable X equals x is defined as the probability of its underlying event E_x occurring:

$$P(X = x) \triangleq P(\{s \in \mathcal{S} \mid X(s) = x\})$$

This is often written in compact shorthand as:

$$P(X = x) = P(s \in \mathcal{S} : X(s) = x)$$

Example: Tossing Two Fair Coins

Let $\mathcal{S} = \{HH, HT, TH, TT\}$ with equally likely outcomes ($P(s) = 1/4$). Let X represent the **number of Heads**:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0$$

To find $P(X = 1)$, we identify which outcomes in $\mathcal{S}$ map to 1:

$$E_1 = \{s \in \mathcal{S} \mid X(s) = 1\} = \{HT, TH\}$$

Since E_1 is a standard, well-defined event in our sample space, we can calculate its probability:

$$P(X = 1) = P(E_1) = P(\{HT, TH\}) = \frac{2}{4} = 0.5$$

The number 1 *is assigned the collective probability mass of the outcomes* HT *and* TH.

Extending to Intervals and Ranges

We do not just calculate probability at single points. We often want to know the probability of a range, such as $P(a \leq X \leq b)$ or $P(X \leq x)$.

The exact same **mapping** rule applies! We collect all outcomes in the sample space $\mathcal{S}$ that map into that interval:

$$P(a \leq X \leq b) \triangleq P(\{s \in \mathcal{S} \mid a \leq X(s) \leq b\})$$

Shorthand Notation Caveat

When you write $P(X \leq x)$, you are using a convenient shorthand.
The mathematical reality behind that shorthand is:

$$P(X \leq x) = P(\{s \in \mathcal{S} \mid X(s) \leq x\})$$

This translation connects the real number world $(X \leq x)$ back to the set theory world $(\mathcal{S})$ where Kolmogorov's Axioms live!

Exercise 1: Cloud Database Reliability

A cloud database replicates data across 3 independent servers. The sample space $\mathcal{S}$ represents the state of each server: (x_1, x_2, x_3) where $x_i \in \{0, 1\}$ ($1 =$ online, $0 =$ offline). The probability of any individual server failing (offline) is 0.1. The servers operate independently.
Let the random variable C represent the **available system throughput** (in Gbps), mapped as a function of the number of online servers $k = x_1 + x_2 + x_3$:

$$C(s) = \begin{cases} 100 & \text{if } k = 3 \\ 50 & \text{if } k = 2 \\ 20 & \text{if } k = 1 \\ 0 & \text{if } k = 0 \end{cases}$$

Calculate the probability $P(C \geq 50)$.

Exercise 2: Reactor Pressure Telemetry

Consider an industrial pressure sensor placed in a chemical reactor. The operational pressure s (in psi) ranges uniformly over the sample space $\mathcal{S} = [0, 200]$ psi, where $P([a, b]) = \frac{b-a}{200}$ for any sub-interval.
The safety telemetry system converts the raw pressure s to a safety index Y using a piecewise logarithmic mapping:

$$Y(s) = \begin{cases} 0 & \text{if } 0 \leq s < 1 \quad \text{(noise floor)} \\ 20 \log_{10}(s) & \text{if } 1 \leq s \leq 100 \quad \text{(telemetry monitoring)} \\ 40 & \text{if } 100 < s \leq 200 \quad \text{(saturation limit)} \end{cases}$$

Your Task

1. Translate the telemetry reading event $\{20 \leq Y \leq 40\}$ back to its corresponding pre-image subset $E \subseteq \mathcal{S}$.
2. Calculate the probability $P(20 \leq Y \leq 40)$ using the probability law.

What is a "Distribution"?

A random variable X maps outcomes from a sample space $\mathcal{S}$ to the real numbers $\mathbb{R}$.
The way the probability mass is scattered, allocated, or "distributed" over these real numbers is called the **probability distribution** of X.

To study distributions systematically, we use specific functions:

- **The Cumulative Distribution Function (CDF)** — Applies universally to all random variables.
- **The Probability Mass Function (PMF)** — Specifically for discrete random variables.
- **The Probability Density Function (PDF)** — Specifically for continuous random variables.

Why Do We Need a Distribution Function?

Previously, calculating probabilities required looking backward to the sample space:

$$P(X \in B) = P(\{s \in \mathcal{S} \mid X(s) \in B\})$$

While mathematically sound, this approach has severe practical limitations:

- It is highly inefficient to map back to the sample space $\mathcal{S}$ for every single probability calculation.
- In many real-world applications, the original sample space $\mathcal{S}$ is complex, high-dimensional, or completely hidden from us.

The Solution

A **distribution function** acts as a mathematical proxy. Once we establish this function on the real line $\mathbb{R}$:

- **All** probabilities associated with X can be calculated using this function **alone**.

The Cumulative Distribution Function (CDF)

The most fundamental and universal way to describe the distribution of any random variable is through cumulative probabilities.

Definition

Cumulative Distribution Function Let X be a random variable. The **Cumulative Distribution Function (CDF)** of X, denoted by $F(x)$ (or $F_X(x)$), is defined for any real number x as:

$$F(x) \triangleq P(X \leq x) \quad \text{for } x \in \mathbb{R}$$

Key Concept

The value $F(x)$ represents the accumulated probability of all outcomes in the sample space that map to a value less than or equal to x.

Why $(-\infty, x]$ is Enough: Reconstructing Events

The CDF is defined strictly using semi-infinite intervals:

$$F(a) \triangleq P(X \leq x) = P(X \in (-\infty, x])$$

A Fundamental Result

Every "nice" or mathematically well-behaved subset of the real numbers can be constructed from intervals of the form $(-\infty, x]$ using basic set operations.

Consequently, the probability of any reasonable event can be written as an addition or subtraction combination of the distribution function values:

Properties of the CDF

The CDF $F(x)$ is a mathematically well-behaved function. For any random variable X, its CDF satisfies the following structural properties:

1. **Monotonicity:** $F(x)$ is a non-decreasing function:

$$\text{If } a \leq b, \text{ then } F(a) \leq F(b)$$

2. **Asymptotic Limits:**

$$\lim_{x \to \infty} F(x) = 1 \quad \text{and} \quad \lim_{x \to -\infty} F(x) = 0$$

3. **Right-Continuity:** $F(x)$ is continuous from the right at every point:

$$\lim_{t \to x^+} F(t) = F(x)$$

Discrete Case: Probability Mass Function (PMF)

For discrete random variables (which take on countable values), we define probability at exact, individual points.

Definition

Probability Mass Function Let X be a discrete random variable with range $\mathcal{T}$. The **Probability Mass Function (PMF)** of X, denoted by $p(x)$ (or $p_X(x)$), is:

$$p(x) \triangleq P(X = x) \quad \text{for } x \in \mathbb{R}$$

Fundamental Axiomatic Properties

Every valid PMF must satisfy:

- $p(x) \geq 0$ for all x.
- $\sum_{x \in \mathcal{T}} p(x) = 1$.

Continuous Case: Probability Density Function (PDF)

For continuous random variables, the probability of taking on any single exact value is zero ($P(X = x) = 0$). Instead, we describe the probability over an interval using density.

Definition

Probability Density Function Let X be a continuous random variable. A non-negative, integrable function $f(x)$ is the **Probability Density Function (PDF)** of X if for any interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x)\,dx$$

Axiomatic Constraints

- $f(x) \geq 0$ for all $x \in \mathbb{R}$.
- $\int_{-\infty}^{\infty} f(x)\,dx = F(\infty) = 1$.

Exercise: Verifying a Valid PMF

Suppose a system database query returns a variable number of matching records, X. The probability mass function of X is modeled by:

$$p(x) = \begin{cases} k \cdot x^2 & \text{if } x \in \{1, 2, 3, 4\} \\ 0 & \text{otherwise} \end{cases}$$

Your Task

1. Find the value of the normalization constant k such that $p(x)$ represents a mathematically valid PMF.
2. Determine and write down the Cumulative Distribution Function (CDF) $F(x)$ for this discrete random variable.

Exercise: Verifying a Valid PDF

Consider a continuous random variable X representing the time (in hours) a user spends actively on a platform in a session. The density is modeled by:

$$f(x) = \begin{cases} c \cdot (2x - x^2) & \text{if } x \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$$

Your Task

Find the value of the normalization constant c such that $f(x)$ represents a mathematically valid PDF, and determine the CDF $F(x)$.

Expectation: The Center of Mass

The **mathematical expectation** (or mean) of a random variable represents its long-run average value over infinitely repeated trials.

Physically, it acts as the **center of gravity** (centroid) of the probability distribution.

Definition

Expected Value (Discrete) Let X be a discrete random variable with support set $\mathcal{T}$ and probability mass function $p(x) = P(X = x)$. The expected value of X, denoted $E[X]$ or μ, is:

$$E[X] \triangleq \sum_{x \in \mathcal{T}} x \cdot p(x)$$

This sum is defined provided that it is absolutely convergent, i.e., $\sum |x| p(x) < \infty$.

Expectation: Continuous Analogue

For a continuous random variable, we cannot sum over individual points because any single point has zero probability.
Instead, we integrate over the continuous support using the probability density function (PDF).

Definition

Expected Value (Continuous) Let X be a continuous random variable with probability density function $f(x)$. The expected value of X is defined as:

$$E[X] \triangleq \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

This integral is defined provided that it is absolutely convergent, i.e., $\int |x| f(x)\, dx < \infty$.

Linearity of Expectation

Expectation is a **linear operator**. This means it preserves scaling and addition, regardless of whether the random variables are dependent or independent.

Theorem

Linearity Properties Let X be a random variable, and let a and b be constants. Then:

$$E[aX + b] = aE[X] + b$$

Proof: Linearity of Expectation

We prove this for the discrete case. The continuous proof follows identically by replacing summations with integrals.
Let X have support $\mathcal{T}$ and PMF $p(x)$. Apply the definition of expectation to $g(X) = aX + b$:

$$\begin{aligned} E[aX + b] &= \sum_{x \in \mathcal{T}} (ax + b) \cdot p(x) \\ &= \sum_{x \in \mathcal{T}} (ax \cdot p(x) + b \cdot p(x)) \\ &= \sum_{x \in \mathcal{T}} ax \cdot p(x) + \sum_{x \in \mathcal{T}} b \cdot p(x) \end{aligned}$$

Pulling the constants out of the summations yields:

$$E[aX + b] = a \sum_{x \in \mathcal{T}} x \cdot p(x) + b \sum_{x \in \mathcal{T}} p(x)$$

Applying $E[X] = \sum xp(x)$ and Axiom 2 ($\sum p(x) = 1$):

$$E[aX + b] = aE[X] + b \cdot (1) = aE[X] + b \quad \blacksquare$$

Variance: Quantifying Spread

While the mean tells us where the distribution is centered, it tells us nothing about how spread out the values are.

Variance measures the average squared deviation of the random variable from its mean (μ).

Definition

Variance Let X be a random variable with mean $\mu = E[X]$. The variance of X, denoted $Var(X)$ or σ^2, is:

$$Var(X) \triangleq E[(X - \mu)^2]$$

Standard Deviation

Because variance is measured in squared units, we often use the **Standard Deviation** (σ) to keep the units consistent with the original data:

Standard Formula for Variance

Evaluating the definition $E[(X-\mu)^2]$ directly can be algebraically tedious. There exists a much simpler, standard formula.

Theorem

For any random variable X:

$$Var(X) = E[X^2] - (E[X])^2$$

Derivation:

Let $\mu = E[X]$. Expanding the squared term inside the expectation definition:

$$Var(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2]$$

Now, apply the linearity of expectation to distribute the operator across each term:

$$Var(X) = E[X^2] - E[2\mu X] + E[\mu^2]$$

Since μ is a constant, we can pull it out of the expectation. Also, the expectation of a constant is the constant itself ($E[\mu^2] = \mu^2$):

$$Var(X) = E[X^2] - 2\mu E[X] + \mu^2$$

Substitute $E[X] = \mu$ back into the expression:

$$Var(X) = E[X^2] - 2\mu(\mu) + \mu^2 = E[X^2] - 2\mu^2 + \mu^2$$

$$Var(X) = E[X^2] - \mu^2 = E[X^2] - (E[X])^2 \quad \blacksquare$$

Properties of Variance: Scaling and Shifts

Unlike expectation, variance is **not** a linear operator. Adding a constant shift does not change the spread of the data, while scaling a variable scales its variance quadratically.

Theorem

Variance Transformations Let X be a random variable, and let a and b be constants. Then:

$$Var(aX + b) = a^2 Var(X)$$

Proof: Variance Transformations

Let $Y = aX + b$. First, find the mean of the transformed variable Y:

$$E[Y] = E[aX + b] = aE[X] + b$$

Now, substitute Y and $E[Y]$ into the definition of variance:

$$\begin{aligned} Var(aX + b) &= E\left[((aX + b) - E[aX + b])^2\right] \\ &= E\left[(aX + b - (aE[X] + b))^2\right] \end{aligned}$$

Notice that the constant shift b cancels out completely:

$$\begin{aligned} Var(aX + b) &= E\left[(aX - aE[X])^2\right] \\ &= E\left[(a(X - E[X]))^2\right] \\ &= E\left[a^2(X - E[X])^2\right] \end{aligned}$$

Pulling the constant factor a^2 out of the expectation:

$$Var(aX + b) = a^2 E\left[(X - E[X])^2\right] = a^2 Var(X) \quad \blacksquare$$

Exercise 1: Server Latency Cost Mapping

An API server experiences variable request latency. Let X represent the processing latency (in milliseconds) with the following PMF:

x	10 ms	50 ms	100 ms
$p(x)$	0.6	0.3	0.1

The operating cost (in micro-dollars) of processing a request of latency X is determined by the quadratic relationship:

$$C(X) = 2X^2 + 10$$

Your Task

Calculate the expected processing cost $E[C(X)]$ per request.

Exercise 2: Precision Sensor Noise

An industrial analog temperature sensor outputs a voltage reading V. Due to thermal interference, the sensor error $X = V - V_{\text{true}}$ (in millivolts) is a continuous random variable distributed uniformly on the range $[-5, 5]$ mV, with a constant probability density:

$$f(x) = \frac{1}{10} \quad \text{for } x \in [-5, 5]$$

Your Task

1. Compute the expected calibration error $E[X]$.
2. Compute the variance $Var(X)$ to quantify the sensor's precision limit.

Central Tendency: The Mathematical Mean

The expected value $E[X]$ we defined is physically known as the **mean** (μ). It is the most common measure of **central tendency**.

Key Characteristics of the Mean

- **Physical Centroid:** It acts as the exact balancing point of the probability mass curve or histogram.
- **Sensitivity:** Because its calculation factors in every value weighted by its probability, the mean is **highly sensitive to outliers** (skewed tails).
- **Linear Balance:** The sum (or integral) of deviations from the mean is always exactly zero:

$$E[X - \mu] = 0$$

Alternative Measures: The Median

When a distribution has extreme values or heavy asymmetry, the mean may not reflect the "typical" value. In such cases, we use the **median**.

Definition

The Median (m) The median m of a random variable X is a real number that divides the probability distribution into two equal halves.

- **Continuous Case:** m is the value satisfying:

$$F(m) = \int_{-\infty}^{m} f(x)\,dx = 0.5$$

- **Discrete Case:** m is any value satisfying:

$$P(X \leq m) \geq 0.5 \quad \text{and} \quad P(X \geq m) \geq 0.5$$

Core Advantage: The median is **robust**; changing extreme values has absolutely zero effect on its position.

Alternative Measures: The Mode

Another fundamental measure of central tendency is the **mode**, which indicates the most likely or popular outcome.

Definition

The Mode (x^*) The mode x^* is the value of x where the probability distribution reaches its maximum concentration or peak.

- **Discrete Case:** The point x^* that maximizes the PMF:

$$p(x^*) \geq p(x) \quad \text{for all } x$$

- **Continuous Case:** The point x^* that maximizes the density function $f(x)$ (typically found where $f'(x^*) = 0$ and $f''(x^*) < 0$).

Distribution Shapes

Unlike mean and median, a distribution can be **unimodal** (one peak), **bimodal** (two peaks), or **multimodal** (several peaks).

Comparing Central Tendency: Symmetry vs. Skewness

The relative positions of the Mean, Median, and Mode reveal the geometric **skewness** (asymmetry) of a distribution.

1. Perfectly Symmetric Distributions

The peaks, physical balance points, and 50% splitting lines align perfectly:

$$\text{Mean} = \text{Median} = \text{Mode}$$

2. Skewed Distributions

- **Right-Skewed (Positive):** A long right tail pulls the mean upward. The mode remains at the main peak, with the median sitting safely in between:

$$\text{Mode} < \text{Median} < \text{Mean}$$

- **Left-Skewed (Negative):** A long left tail pulls the mean downward:

$$\text{Mean} < \text{Median} < \text{Mode}$$

Robust Variability: Median Absolute Deviation

Just as the mean has a robust counterpart (the median), variance has a robust alternative known as the **Median Absolute Deviation (MAD)**. Variance and standard deviation are highly sensitive to extreme outliers because they square deviations. MAD resolves this.

Definition

Median Absolute Deviation (MAD) Let X be a random variable with median m. The Median Absolute Deviation of X is defined as:

$$MAD \triangleq \text{median}(|X - m|)$$

Why Use MAD?

- **Extreme Outlier Resistance:** A single outlier can inflate variance infinitely, whereas it barely changes the MAD.
- **Preserves Scale:** It measures variability in the same units as the original data without squaring the errors.

Why Do We Need Joint Distributions?

In most real-world scenarios, an experiment produces multiple numerical outcomes simultaneously.
To model relationships, associations, and dependencies, we must look at variables collectively rather than in isolation.

Key Objectives

- Learn to describe the joint behavior of multiple random variables.
- Understand how to extract individual probability laws from joint laws.
- Quantify the strength and direction of linear relationships between variables.

Discrete Case: Joint Probability Mass Function

When X and Y are both discrete random variables, we describe their joint probabilities using a single function of two variables.

Definition

Joint Probability Mass Function The joint probability mass function (joint PMF) of discrete random variables X and Y is defined for all real numbers x and y as:

$$p(x, y) \triangleq P(X = x, Y = y)$$

Axiomatic Properties

Every valid joint PMF must satisfy:

- $p(x, y) \geq 0$ for all (x, y).
- $\sum_x \sum_y p(x, y) = 1$.

Continuous Case: Joint Probability Density Function

When variables are continuous, probability at a single point is zero. Instead, we define probability over regions using a joint density surface.

Definition

Joint Probability Density Function The joint probability density function (joint PDF) of continuous random variables X and Y is a function $f(x, y)$ such that for any region $A \subseteq \mathbb{R}^2$:

$$P((X, Y) \in A) = \iint_A f(x, y)\, dx\, dy$$

Required Constraints

- $f(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$.
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$.

Extracting Individual Laws: Marginal Distributions

If we are given a joint distribution, we can recover the individual probability law of X or Y by summing or integrating out the other variable.

Marginal PMF (Discrete)

We sum across all possible values of the unwanted variable:

$$p_X(x) = \sum_y p(x, y) \quad \text{and} \quad p_Y(y) = \sum_x p(x, y)$$

Marginal PDF (Continuous)

We integrate across the entire domain of the unwanted variable:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

Example: Discrete Joint PMF

Let X and Y be discrete random variables with the following joint PMF table:

$X \setminus Y$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.1	0.2	0.1
$x = 1$	0.2	0.3	0.1

Finding the Marginal PMF of X

To find $p_X(x)$, we sum across the rows (integrating out Y):

- $p_X(0) = \sum_{y=0}^{2} p(0, y) = 0.1 + 0.2 + 0.1 = 0.4$
- $p_X(1) = \sum_{y=0}^{2} p(1, y) = 0.2 + 0.3 + 0.1 = 0.6$

Sanity Check: $\sum_x p_X(x) = 0.4 + 0.6 = 1$.

Example: Continuous Joint PDF

Let X and Y have the following joint density function:

$$f(x,y) = \begin{cases} c(x+2y), & 0 < x < 1,\ 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 1: Find the normalizing constant c

We use the total probability constraint $\int\int f(x,y)\,dx\,dy = 1$:

$$\begin{aligned} 1 &= \int_0^1 \int_0^1 c(x+2y)\,dx\,dy \\ &= c\int_0^1 \left[\frac{x^2}{2} + 2xy\right]_{x=0}^{x=1} dy \\ &= c\int_0^1 \left(\frac{1}{2} + 2y\right) dy \\ &= c\left[\frac{y}{2} + y^2\right]_0^1 = c\left(\frac{1}{2} + 1\right) = \frac{3}{2}c \end{aligned}$$

Example: Continuous Marginal PDF

Continuing with our normalized joint PDF: $f(x,y) = \frac{2}{3}(x+2y)$ for $0 < x < 1$ and $0 < y < 1$.

Step 2: Find the marginal PDF of X

To find $f_X(x)$, we integrate out Y over its entire domain:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x,y)\,dy \\ &= \int_0^1 \frac{2}{3}(x+2y)\,dy \\ &= \frac{2}{3}\left[xy + y^2\right]_{y=0}^{y=1} \\ &= \frac{2}{3}(x+1) \end{aligned}$$

Therefore, $f_X(x) = \frac{2}{3}(x+1)$ for $0 < x < 1$, and 0 otherwise.

Independence of Random Variables

We previously defined independence for events. Similarly, two random variables are independent if knowing the value of one provides zero information about the other.

Definition

Statistical Independence Random variables X and Y are independent if their joint distribution factors cleanly into the product of their individual marginal distributions.

- **Discrete Case:**

$$p(x, y) = p_X(x) \cdot p_Y(y) \quad \text{for all } x, y$$

- **Continuous Case:**

$$f(x, y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y$$

Linearity of Expectation

Expectation is a **linear operator**. This means it preserves scaling and addition, regardless of whether the random variables are dependent or independent.

Theorem

Linearity Properties Let X and Y be two jointly distributed random variables. Then:

$$E[X + Y] = E[X] + E[Y]$$

Proof: Linearity of Expectation

To fully establish linearity, we must show that expectation distributes over addition for **any** two random variables X and Y, without assuming independence. Let X and Y have joint support $\mathcal{T}_{X,Y}$ and joint PMF $p(x, y)$. Apply the definition of expectation to $g(X, Y) = X + Y$:

$$\begin{aligned} E[X+Y] &= \sum_{(x,y)\in\mathcal{T}_{X,Y}} (x+y)\cdot p(x,y) \\ &= \sum_{x}\sum_{y} (x\cdot p(x,y) + y\cdot p(x,y)) \\ &= \sum_{x}\sum_{y} x\cdot p(x,y) + \sum_{y}\sum_{x} y\cdot p(x,y) \end{aligned}$$

Factoring out the variables that do not depend on the inner summation index:

$$E[X+Y] = \sum_{x} x\left(\sum_{y} p(x,y)\right) + \sum_{y} y\left(\sum_{x} p(x,y)\right)$$

$$E[X+Y] = \sum_{x} x\cdot p_X(x) + \sum_{y} y\cdot p_Y(y) = E[X] + E[Y] \quad \blacksquare$$

Covariance: Quantifying Relationship Direction

While expectation tells us about the center of individual variables, we need a metric to evaluate how they vary together.

Definition

Covariance Let X and Y be random variables with means $\mu_X = E[X]$ and $\mu_Y = E[Y]$. The covariance of X and Y, denoted $Cov(X, Y)$, is:

$$Cov(X, Y) \triangleq E[(X - \mu_X)(Y - \mu_Y)]$$

Intuitive Interpretation

- **Positive Covariance:** When X is above its mean, Y tends to be above its mean (positive linear trend).
- **Negative Covariance:** When X is above its mean, Y tends to be below its mean (inverse linear trend).
- **Zero Covariance:** No linear association.

Covariance

Evaluating the definition $E[(X - \mu_X)(Y - \mu_Y)]$ directly requires calculating deviations for every point, which is algebraically tedious. Instead, we use a much simpler standard identity.

Theorem

For any two random variables X and Y:

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

Let us derive this relationship on the next slide using the properties of expectation.

Derivation:

Let $\mu_X = E[X]$ and $\mu_Y = E[Y]$. Expanding the terms inside the expectation definition:

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$Cov(X, Y) = E[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y]$$

Apply the linearity of expectation to distribute the operator across each term:

$$Cov(X, Y) = E[XY] - E[\mu_Y X] - E[\mu_X Y] + E[\mu_X \mu_Y]$$

Since the means μ_X and μ_Y are constants, we pull them out of the expectation:

$$Cov(X, Y) = E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$$

Substitute $E[X] = \mu_X$ and $E[Y] = \mu_Y$ back into the expression:

$$Cov(X, Y) = E[XY] - \mu_Y \mu_X - \mu_X \mu_Y + \mu_X \mu_Y$$

$$Cov(X, Y) = E[XY] - \mu_X \mu_Y = E[XY] - E[X]E[Y] \quad \blacksquare$$

Properties of Covariance

Covariance behaves according to the following mathematical rules:

Core Covariance Properties

Let X, Y, and Z be random variables, and let a and b be constants.

1. **Symmetry:** $Cov(X, Y) = Cov(Y, X)$
2. **Covariance with Self is Variance:** $Cov(X, X) = Var(X)$
3. **Scaling and Bilinearity:** $Cov(aX, bY) = ab\, Cov(X, Y)$
4. **Additivity:** $Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$

Variance is NOT Linear

Unlike expectation, variance is a **quadratic** operator. Because it measures squared deviations, it does not simply preserve addition or scaling.

The General Rule for Variance

Let X and Y be any two random variables, and let $a, b \in \mathbb{R}$ be constants. The variance of their linear combination is:

$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab\, Cov(X, Y)$$

If we simply add the variables together ($a = 1, b = 1$), this reduces to the fundamental sum formula:

$$Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

Key Takeaway: $Var(X + Y) \neq Var(X) + Var(Y)$ unless the variables are completely uncorrelated (i.e., $Cov(X, Y) = 0$).

Proof: Variance of a Sum

Let us formally prove that $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$. By definition, $Var(Z) = E[(Z - E[Z])^2]$. Let $Z = X + Y$. Applying our rule of linear expectation, $E[X+Y] = E[X] + E[Y]$.

Algebraic Expansion

Group the random variables directly with their respective population means:

$$\begin{aligned} Var(X+Y) &= E\left[\Big((X+Y) - (E[X] + E[Y])\Big)^2\right] \\ &= E\left[\Big((X - E[X]) + (Y - E[Y])\Big)^2\right] \end{aligned}$$

Expand to get:

$$= E\left[(X - E[X])^2 + 2(X - E[X])(Y - E[Y]) + (Y - E[Y])^2\right]$$

$$\implies Var(X+Y) = Var(X) + 2Cov(X,Y) + Var(Y)$$

Independence and Covariance

What happens to covariance if the random variables are independent?

Theorem

Covariance of Independent Variables If X and Y are independent random variables, then:

$$Cov(X, Y) = 0$$

Proof Sketch

If X and Y are independent, then their joint expectation factors: $E[XY] = E[X]E[Y]$. Substituting this into our computational formula:

$$Cov(X, Y) = E[XY] - E[X]E[Y] = E[X]E[Y] - E[X]E[Y] = 0 \quad \blacksquare$$

The Converse is not true

The converse is **not** generally true. If $Cov(X, Y) = 0$, the variables are uncorrelated, but they are **not necessarily independent**.

Exercise 1: Discrete Covariance Calculation

Let X and Y be discrete random variables representing the outcome of a simple system. Their joint PMF, $p(x, y)$, is defined by the following table:

	$Y = 1$	$Y = 2$
$X = 0$	0.1	0.4
$X = 1$	0.3	0.2

Your Task

1. Find the marginal PMFs of X and Y.
2. Compute the expected values $E[X]$ and $E[Y]$.
3. Compute the covariance $Cov(X, Y)$.

Exercise 2: Continuous Covariance Calculation

Let X and Y be continuous random variables. Their joint density function is defined as:

$$f(x,y) = \begin{cases} x+y & \text{if } 0 \le x \le 1 \text{ and } 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Your Task

1. Compute the expected values $E[X]$ and $E[Y]$.
2. Compute the expected product $E[XY]$.
3. Compute the covariance $Cov(X,Y)$.

The Cauchy-Schwarz Inequality

The relationship between the expectations of products and individual second moments is bounded by a fundamental mathematical law.

Theorem

Cauchy-Schwarz Inequality For any two random variables X and Y with finite variances:

$$(E[XY])^2 \leq E[X^2]E[Y^2]$$

In terms of deviations from their respective means, this is equivalent to:

$$(Cov(X, Y))^2 \leq Var(X)Var(Y)$$

Physical Implication

This inequality guarantees that covariance cannot grow infinitely large. It is bounded strictly by the individual spreads (variances) of the two variables, which allows us to define a standardized scale for relationships.

Sketch of Proof

The proof relies on the property that the expectation of a non-negative random variable is always non-negative.

1. Starting Point

For any constant $a \in \mathbb{R}$, consider the squared difference $E[(aX - bY)^2] \geq 0$.

2. Expansion

Expanding the quadratic inside the expectation:

$$E[a^2X^2 - 2aXY + Y^2] \geq 0$$

By the linearity of expectation, we obtain:

$$a^2E[X^2] - 2aE[XY] + E[Y^2] \geq 0$$

Sketch of Proof

3. The Discriminant Condition

Treating this as a quadratic in a , for the expression to be ≥ 0 for all a, the discriminant must be non-positive ($\Delta \leq 0$):

$$(2E[XY])^2 - 4E[X^2]E[Y^2] \leq 0$$

$$4(E[XY])^2 \leq 4E[X^2]E[Y^2]$$

4. Result

Dividing by 4 yields the Cauchy-Schwarz Inequality:

$$(E[XY])^2 \leq E[X^2]E[Y^2]$$

5. Equality Condition

The equality holds iff there exists an a such that $aX = Y$ wp 1.

The Correlation Coefficient: Standardizing Covariance

While covariance tells us the direction of a linear relationship, its numerical value depends on the arbitrary units of X and Y. To fix this, we normalize it.

Definition

Correlation Coefficient The correlation coefficient of two random variables X and Y, denoted by $\rho(X, Y)$ (or simply ρ), is defined as:

$$\rho(X, Y) \triangleq \frac{Cov(X, Y)}{\sigma_X \sigma_Y} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

Properties of Correlation

By direct application of the Cauchy-Schwarz inequality:

- **Boundedness:** $-1 \leq \rho(X, Y) \leq 1$.
- **Perfect Linear Relationship:** $\rho = 1$ indicates a perfect positive linear trend; $\rho = -1$ indicates a perfect negative linear trend.

Skewness

For a continuous random variable X with PDF $f(x)$, mean μ, and standard deviation σ, the skewness is defined via the integral:

$$\gamma_1 = \frac{1}{\sigma^3} \int_{-\infty}^{\infty} (x - \mu)^3 f(x)\, dx$$

Proof of Zero Skewness for Symmetry

Assume $f(x)$ is perfectly symmetric about μ, meaning $f(\mu - y) = f(\mu + y)$. We can shift our axis by letting $y = x - \mu$:

$$\int_{-\infty}^{\infty} y^3 f(\mu + y)\, dy$$

Notice that y^3 is an **odd** function ($(-y)^3 = -y^3$), while our shifted PDF $f(\mu + y)$ is an **even** function. The product of an odd and even function is odd. Integrating an odd function over a symmetric interval $[-\infty, \infty]$ perfectly cancels out, yielding 0.

Rethinking Kurtosis: The Variance of Deviations

To understand why thicker tails increase kurtosis, let us standardize our variable: $Z = \frac{X-\mu}{\sigma}$. By definition, $E[Z^2] = 1$ and Kurtosis is $E[Z^4]$.Let us examine the variance of these squared deviations, $Var(Z^2)$, using the standard variance expansion $Var(Y) = E[Y^2] - (E[Y])^2$:

$$Var(Z^2) = E[(Z^2)^2] - (E[Z^2])^2$$

$$Var(Z^2) = E[Z^4] - (1)^2$$

Rearranging this gives us some insight into Kurtosis:

$$\text{Kurtosis} = Var(Z^2) + 1$$

It shows that Kurtosis is entirely driven by how much the **squared deviations vary**.

Why Thicker Tails Drive Kurtosis Higher

The relationship Kurtosis $= 1 + Var(Z^2)$ perfectly explains the geometric behavior of the tails.

1. Thick Tails

Z^2 represents the squared distance of a point from the mean. If a distribution has **thick tails**, it possesses a higher probability of generating large values producing occasionally massive values for Z^2.

2. Conclusion

Thicker tails $\implies$ larger extreme Z^2 values $\implies$ a massive increase in $Var(Z^2)$. Because Kurtosis is simply $Var(Z^2) + 1$, adding mass to the tails forces the kurtosis higher.

Standard Discrete Distributions: The Discrete Uniform

We begin our exploration of standard distributions with the most fundamental concept of fairness: a system where every outcome in a finite set is equally likely.

The Discrete Uniform Distribution

Let a random variable X take on any integer value in a contiguous range from a to b inclusive.

- The total number of possible outcomes is $N = b - a + 1$.
- Because no outcome is favored, the probability is distributed evenly across the entire support.

Probability Mass Function (PMF)

We write $X \sim \text{DiscreteUniform}(a, b)$. The PMF is:

$$P(X = k) = \frac{1}{N} = \frac{1}{b - a + 1}, \quad \text{for } k = a, a + 1, \ldots, b$$

Discrete Uniform: Expectation and Variance

1. Expectation (The Midpoint)

Using the definition of expectation and factoring out the constant probability:

$$E[X] = \sum_{k=a}^{b} k \frac{1}{N} = \frac{1}{N} \sum_{k=a}^{b} k$$

The sum of this arithmetic sequence is the average of the first and last terms multiplied by the total number of terms (N):

$$E[X] = \frac{1}{N} \left(N \frac{a+b}{2} \right) = \frac{a+b}{2}$$

Discrete Uniform Variance

Calculating the sum of squares directly from a to b is messy. Instead, we define a shifted variable $Y = X - a + 1$.
Y now takes values $1, 2, \ldots, N$. Because shifting a distribution does not change its spread, $Var(X) = Var(Y)$.

1. Find the Second Moment $E[Y^2]$

Using the standard formula for the sum of consecutive squares:

$$E[Y^2] = \sum_{k=1}^{N} k^2 \frac{1}{N} = \frac{1}{N}\left[\frac{N(N+1)(2N+1)}{6}\right] = \frac{(N+1)(2N+1)}{6}$$

2. Calculate Final Variance

$$Var(Y) = \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2}\right)^2$$

$$\implies Var(Y) = \frac{N+1}{2}\left[\frac{N-1}{6}\right] = \frac{N^2-1}{12} \quad \blacksquare$$

The Continuous Uniform Distribution

We begin with the continuous equivalent of perfect fairness.

The Definition

A random variable $X \sim \text{Uniform}(a, b)$ is equally likely to take any real value in the continuous interval $[a, b]$. Because the density is evenly spread out, the PDF is a perfectly flat, horizontal line.

The PDF Formula

To ensure the total area under the curve equals exactly 1 (a rectangle of width $b - a$), the height must be $\frac{1}{b-a}$:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Continuous Uniform: Expected Value

Let $X \sim \text{Uniform}(a, b)$ with PDF $f(x) = \frac{1}{b-a}$. We find the expected value via integration:

$$\begin{aligned} E[X] &= \int_a^b x f(x) dx \\ &= \int_a^b x \left(\frac{1}{b-a} \right) dx \\ &= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b \\ &= \frac{1}{b-a} \left(\frac{b^2 - a^2}{2} \right) \end{aligned}$$

Factoring the difference of squares in the numerator, $b^2 - a^2 = (b-a)(b+a)$:

$$E[X] = \frac{1}{b-a} \frac{(b-a)(b+a)}{2} = \frac{a+b}{2} \quad \blacksquare$$

Continuous Uniform: Variance

First, we calculate the second moment, $E[X^2]$:

$$E[X^2] = \int_a^b x^2 \left(\frac{1}{b-a}\right) dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b$$
$$= \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(a^2+ab+b^2)}{3(b-a)} = \frac{a^2+ab+b^2}{3}$$

Now, apply the variance formula $Var(X) = E[X^2] - (E[X])^2$:

$$Var(X) = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2$$
$$= \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4}$$

Finding a common denominator of 12:

$$Var(X) = \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12}$$
$$= \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12} \quad \blacksquare$$

Generating Random Numbers

Computers are inherently deterministic, meaning they cannot generate truly random numbers. Instead, we rely on algorithms called **Pseudo-Random Number Generators (PRNGs)** to create sequences $U_1, U_2, \ldots$ that behave like standard uniforms, Uniform$(0, 1)$.

What must a PRNG ensure?

To be mathematically valid for simulation, the algorithm must guarantee:

- **Uniformity:** The numbers must be spread evenly across $[0, 1]$. No sub-interval should be favored.
- **Independence:** Successive numbers must lack correlation. Knowing U_n should give you no predictable information about U_{n+1}.
- **Long Period:** Because the system has finite memory and it is determeinistic, the sequence will eventually repeat. The length of this cycle (the "period") must be massive to avoid repeating patterns during a simulation.
- **Efficiency:** It must be computationally fast.

A Simple Algorithm: The Linear Congruential Generator

One of the oldest and most fundamental PRNG algorithms is the **Linear Congruential Generator (LCG)**. It generates a sequence of pseudo-random integers $X_0, X_1, X_2, \ldots$ using a simple recursive formula:

$$X_{n+1} = (aX_n + c) \pmod{m}$$

Where the algorithm is defined by four integer parameters:

- $m > 0$: the **modulus** (which dictates the maximum possible period)
- $a \in \{1, \ldots, m-1\}$: the **multiplier**
- $c \in \{0, \ldots, m-1\}$: the **increment**
- $X_0 \in \{0, \ldots, m-1\}$: the **seed** (the initial starting state)

To obtain the standard uniform output $U \sim \text{Uniform}(0, 1)$, we scale the result:

$$U_{n+1} = \frac{X_{n+1}}{m}$$

LCG: A "Toy" Example

Let's build a miniature LCG to see the algorithm in action.

The Setup

Let Modulus $m = 9$, Multiplier $a = 2$, Increment $c = 0$, and Seed $X_0 = 1$. The recurrence relation is: $X_{n+1} = (2X_n) \pmod 9$

- $X_1 = (2 \times 1) \pmod 9 = 2 \implies \mathbf{U_1 = 2/9 \approx 0.222}$
- $X_2 = (2 \times 2) \pmod 9 = 4 \implies \mathbf{U_2 = 4/9 \approx 0.444}$
- $X_3 = (2 \times 4) \pmod 9 = 8 \implies \mathbf{U_3 = 8/9 \approx 0.889}$
- $X_4 = (2 \times 8) \pmod 9 = 7 \implies \mathbf{U_4 = 7/9 \approx 0.778}$
- $X_5 = (2 \times 7) \pmod 9 = 5 \implies \mathbf{U_5 = 5/9 \approx 0.556}$
- $X_6 = (2 \times 5) \pmod 9 = 1 \implies \mathbf{U_6 = 1/9 \approx 0.111}$

Note: Modern algorithms use highly complex bitwise operations and massive prime moduli to achieve periods like $2^{19937} - 1$, satisfying the strict criteria from our first slide.

Simulating General Uniform Distributions

How do we get from $U \sim \text{Uniform}(0, 1)$ to other uniform distributions? We use simple linear transformations.

1. Continuous Uniform on $[a, b]$

To generate $X \sim \text{Uniform}(a, b)$, we scale U by the width of our target interval $(b - a)$, and then shift it by the starting point a:

$$X = a + (b - a)U$$

2. Discrete Uniform on $\{a, a + 1, \ldots, b\}$

To generate a random integer uniformly, we first create a continuous uniform variable over a slightly wider range $[a, b + 1)$, and then apply the floor function $\lfloor \cdot \rfloor$ to round down to the nearest integer:

$$X = \lfloor a + (b - a + 1)U \rfloor$$

Notice that the multiplier $(b - a + 1)$ is exactly N, the total number of distinct outcomes.

Generating from any CDF: The Inverse Transform Method

From ChatGPT

What if we want to simulate a random variable X from a distribution with a known Cumulative Distribution Function (CDF), $F(x)$?

The Generalized Inverse Transform Theorem

A valid CDF is non-decreasing but may have flat regions where a standard inverse is undefined. Instead, we use the **generalized (left) inverse**:

$$F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}$$

Let $U \sim \text{Uniform}(0, 1)$. If we define a new random variable as:

$$X = F^{-1}(U)$$

Then X is distributed exactly according to $F(x)$.

Generating from any CDF: The Inverse Transform Method

Proof: We want to show that the CDF of our new variable X is $F(x)$:

$$P(X \leq x) = P(F^{-1}(U) \leq x)$$

Because $F(x)$ is non-decreasing and right-continuous, the generalized inverse satisfies the fundamental equivalence $F^{-1}(u) \leq x \iff u \leq F(x)$. Therefore, we can rewrite the probability:

$$P(F^{-1}(U) \leq x) = P(U \leq F(x))$$

Since $U \sim \text{Uniform}(0, 1)$, we know $P(U \leq c) = c$ for any constant $c \in [0, 1]$. Because a valid CDF guarantees $0 \leq F(x) \leq 1$, it follows that:

$$P(U \leq F(x)) = F(x) \quad \blacksquare$$

Inverse Transform Method: Example

Example: The Exponential Distribution

Suppose X has CDF: $F(x) = 1 - e^{-\lambda x}, \quad \text{for } x \geq 0$

1. Set a standard uniform variable U equal to the CDF:

$$U = 1 - e^{-\lambda X}$$

2. Solve algebraically for X to find the inverse $F^{-1}(U)$:

$$e^{-\lambda X} = 1 - U \implies -\lambda X = \ln(1 - U) \implies X = -\frac{1}{\lambda} \ln(1 - U)$$

3. Because $U \sim \text{Uniform}(0, 1)$, the quantity $(1 - U)$ is also exactly distributed as Uniform$(0, 1)$. For computational efficiency, we often simplify the algorithm to:

$$X = -\frac{1}{\lambda} \ln(U)$$

Inverse Transform Method: Discrete Example

Example: A Generic Discrete Distribution

Suppose X takes positive integer values $i = 1, 2, 3, \ldots$ with probabilities $p_i = P(X = i)$. The CDF is a step function:

$$F(k) = P(X \leq k) = \sum_{i=1}^{k} p_i$$

Because the CDF is a step function, we rely entirely on the generalized inverse $F^{-1}(U) = \inf\{x : F(x) \geq U\}$. This essentially finds the first "step" of the CDF that reaches or exceeds U.

1. Generate a standard uniform variable $U \sim \text{Uniform}(0, 1)$.
2. Accumulate the probabilities $p_1, p_2, \ldots$ until the sum is greater than or equal to U. In other words, find the smallest integer k such that:

$$U \leq \sum_{i=1}^{k} p_i$$

Inverse Transform Method: Discrete Example

A Concrete Discrete Distribution

Suppose X takes three values with the following probabilities (p_i) and CDF values $F(x) = P(X \leq x)$:

- $P(X = 1) = 0.2 \implies F(1) = 0.2$
- $P(X = 2) = 0.5 \implies F(2) = 0.2 + 0.5 = 0.7$
- $P(X = 3) = 0.3 \implies F(3) = 0.7 + 0.3 = 1.0$

Finding the Generalized Inverse $F^{-1}(u)$

Recall $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. To find the inverse, we see where a given probability $u \in (0, 1]$ lands in our CDF steps:

$$F^{-1}(u) = \begin{cases} 1 & \text{if } 0 < u \leq 0.2 \\ 2 & \text{if } 0.2 < u \leq 0.7 \\ 3 & \text{if } 0.7 < u \leq 1.0 \end{cases}$$

The Bernoulli Trial

The Bernoulli Trial

A Bernoulli trial is a random experiment with only two possible outcomes, typically labeled as "Success" (1) and "Failure" (0).

- Let p be the probability of success, where $0 \leq p \leq 1$.
- The probability of failure is therefore $1 - p$ (often denoted as q).

Examples:

- Flipping a biased coin (Heads $= 1$, Tails $= 0$).
- Testing a manufactured component (Pass $= 1$, Defective $= 0$).
- A bit transmitted over a noisy channel (Received correctly $= 1$, Error $= 0$).

The Bernoulli Distribution

If a random variable X represents the outcome of a Bernoulli trial, we write $X \sim \text{Bernoulli}(p)$.
The PMF can be compactly written for $x \in \{0, 1\}$ as:

$$P(X = x) = p^x(1 - p)^{1-x}$$

Using our fundamental definitions of expectation and variance:

- **Expectation:**

$$E[X] = \sum_{x \in \{0,1\}} x \cdot P(X = x) = (1 \cdot p) + (0 \cdot (1 - p)) = p$$

- **Second Moment:**

$$E[X^2] = (1^2 \cdot p) + (0^2 \cdot (1 - p)) = p$$

- **Variance:**

$$Var(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$$

Standard Discrete Models: The Binomial Distribution

What happens if we repeat a Bernoulli trial multiple times?

Definition via Summation

Let $X_1, X_2, \ldots, X_n$ be n **independent** random variables, where each $X_i \sim \text{Bernoulli}(p)$.
We define a new random variable Y as their sum:

$$Y = \sum_{i=1}^{n} X_i$$

Y represents the total number of "Successes" in n independent trials. We say $Y \sim \text{Binomial}(n, p)$.

Examples:

- The number of Heads in 10 coin flips.
- The number of defective items in a batch of 50.

Binomial Probability Mass Function

To find the PMF of Y, we want the probability of getting exactly k successes (where $k \in \{0, 1, \ldots, n\}$).

Deriving the Probability

1. **Specific Sequence:** The probability of getting k successes followed exactly by $n - k$ failures is $p^k(1-p)^{n-k}$.
2. **Counting Combinations:** There are many ways to arrange k successes among n trials. The number of such unique sequences is given by the binomial coefficient $\binom{n}{k}$.

Multiplying these together yields the PMF:

$$P(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Binomial Mean and Variance

We could calculate the expectation by solving $\sum_{k=0}^{n} k\binom{n}{k}p^k(1-p)^{n-k}$, which requires complex algebra.
Instead, we use the linear properties we proved earlier!

1. Expectation (Linearity)

Because expectation is a linear operator:

$$E[Y] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} p = np$$

2. Variance (Independence)

Because the trials X_i are **independent**, the covariance terms vanish, allowing us to distribute the variance operator:

$$Var(Y) = Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) = \sum_{i=1}^{n} p(1-p) = np(1-p)$$

Binomial Expectation:

Let $Y \sim \text{Binomial}(n, p)$ and $q = 1 - p$. We calculate $E[Y]$ directly:

$$E[Y] = \sum_{k=0}^{n} k \binom{n}{k} p^k q^{n-k} = \sum_{k=1}^{n} k \frac{n!}{k!(n-k)!} p^k q^{n-k}$$

Cancel k with $k!$, leaving $(k-1)!$ in the denominator:

$$E[Y] = \sum_{k=1}^{n} \frac{n!}{(k-1)!(n-k)!} p^k q^{n-k}$$

Factor out np to reconstruct a new binomial coefficient:

$$E[Y] = np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} q^{n-k} = np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} q^{(n-1)-(k-1)}$$

Let $j = k - 1$. By the Binomial Theorem, the remaining sum is $(p+q)^{n-1} = 1$:

$$E[Y] = np \cdot 1 = np \quad \blacksquare$$

Binomial Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[Y(Y-1)]$:

$$E[Y(Y-1)] = \sum_{k=0}^{n} k(k-1)\binom{n}{k} p^k q^{n-k}$$

The terms for $k = 0$ and $k = 1$ are zero. Starting at $k = 2$, cancel $k(k-1)$ with $k!$:

$$E[Y(Y-1)] = \sum_{k=2}^{n} \frac{n!}{(k-2)!(n-k)!} p^k q^{n-k}$$

Factor out $n(n-1)p^2$ to rebuild the combinations:

$$E[Y(Y-1)] = n(n-1)p^2 \sum_{k=2}^{n} \frac{(n-2)!}{(k-2)!(n-k)!} p^{k-2} q^{n-k}$$

Just as before, the remaining sum is exactly $(p+q)^{n-2} = 1$. Thus:

$$E[Y(Y-1)] = n(n-1)p^2$$

Binomial Variance:

We just found $E[Y(Y-1)] = n(n-1)p^2$ and we know $E[Y] = np$.

1. Find the Second Moment $E[Y^2]$

Expand the factorial moment: $E[Y(Y-1)] = E[Y^2 - Y] = E[Y^2] - E[Y]$.

Rearranging gives:

$$\begin{aligned} E[Y^2] &= E[Y(Y-1)] + E[Y] \\ &= n(n-1)p^2 + np \\ &= n^2p^2 - np^2 + np \end{aligned}$$

2. Calculate Final Variance

Apply the standard variance formula $Var(Y) = E[Y^2] - (E[Y])^2$:

$$\begin{aligned} Var(Y) &= (n^2p^2 - np^2 + np) - (np)^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= np - np^2 = np(1-p) \quad \blacksquare \end{aligned}$$

The Sample Proportion ($\hat{p}$)

Instead of looking at the total successes Y, let's examine the **average number of successes** per trial, known as the sample proportion: $\hat{p} = \frac{Y}{n}$.

1. Expectation

$$E\left[\frac{Y}{n}\right] = \frac{1}{n}E[Y] = \frac{1}{n}(np) = p$$

The expected proportion of successes is exactly the probability of a single success.

2. Variance

Recall that pulling a constant out of a variance operator squares the constant:

$$Var\left(\frac{Y}{n}\right) = \frac{1}{n^2}Var(Y) = \frac{1}{n^2}np(1-p) = \frac{p(1-p)}{n}$$

The Sample Proportion: As $n \to \infty$

We just established that the variance of the sample proportion is inversely proportional to the number of trials:

$$Var(\hat{p}) = \frac{p(1-p)}{n}$$

The Infinite Limit

What happens mathematically if we repeat the trial an infinite number of times?

$$\lim_{n\to\infty} Var(\hat{p}) = \lim_{n\to\infty} \frac{p(1-p)}{n} = 0$$

Physical Implication: As n grows indefinitely, the sample proportion $\hat{p}$ stops fluctuating and collapses precisely onto the true, underlying probability p. This validates the frequentist definition of probability.

Sums of Independent Random Variables

Let X and Y be two **independent** random variables defined on the same probability space. We are interested in finding the probability distribution of their sum:

$$Z = X + Y$$

To determine the distribution of Z, we analyze all possible pairs (x, y) such that $x + y = z$. Because Y is constrained by $Y = z - X$, independence allows us to determine the distribution using the mathematical operation of **convolution**.

The Discrete Case: PMF Convolution

If X and Y are discrete independent random variables with PMFs $p_X(x)$ and $p_Y(y)$, the PMF of Z is obtained by summing over all valid disjoint supports:

$$p_Z(z) = P(X + Y = z) = \sum_x p_X(x) p_Y(z - x)$$

This operation is frequently denoted as $p_Z = p_X * p_Y$.

Convolution of Continuous Random Variables

The Continuous Case: PDF Convolution

If X and Y are continuous independent random variables characterized by the PDFs $f_X(x)$ and $f_Y(y)$, the density of their sum Z is evaluated via integration:

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx$$

Property: Commutativity

Because algebraic addition is commutative ($X + Y = Y + X$), the convolution operator is symmetric. Thus, the order of the functions under the integral can be inverted:

$$\int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\, dx = \int_{-\infty}^{\infty} f_Y(y) f_X(z-y)\, dy$$

Example: Discrete Support (Geometric Sum)

Problem Setup

Let X and Y be independent Geometric random variables representing the number of trials until the first success, both with success probability p. Their PMF for $k \in \{1, 2, 3, \dots\}$ is:

$$p_X(k) = p_Y(k) = (1-p)^{k-1}p$$

Find the PMF of $Z = X + Y$.

Applying the Convolution Formula

Since the minimum value for both X and Y is 1, the minimum value for Z is $1 + 1 = 2$.

$$p_Z(z) = \sum_x p_X(x)p_Y(z-x)$$

The lower limit for x is 1. For $p_Y(z-x)$ to be non-zero, we also need $z - x \geq 1 \implies x \leq z - 1$.

$$\sum_{x=1}^{z-1}$$

Example: Geometric Sum (Solution)

Simplifying the Summation

We combine the terms with base $(1-p)$ and group the p terms:

$$p_Z(z) = \sum_{x=1}^{z-1} p^2(1-p)^{x-1+z-x-1}$$

$$p_Z(z) = \sum_{x=1}^{z-1} p^2(1-p)^{z-2}$$

Final Result

Since the term inside the sum does not depend on x, we are simply adding a constant to itself $z-1$ times:

$$p_Z(z) = (z-1)p^2(1-p)^{z-2}, \quad \text{for } z = 2, 3, 4, \ldots$$

Note: This is exactly the Negative Binomial distribution NB(2, p), which models

The "Rare Event" Limit

What happens to a Binomial distribution when the number of trials n becomes extremely large, but the probability of success p is extremely small?

The Setup

Consider modeling rare events in a massive population, such as:

- The number of typos in a 500-page book.
- The number of radioactive decays in a millisecond.

Let us take the limit as $n \to \infty$ and $p \to 0$. To keep the mathematics meaningful, we hold their product constant: let $np = \lambda$.
Here, λ represents the **average rate** of successes.

Deriving the Poisson PMF: Setup

Let $Y \sim \text{Binomial}(n, p)$. We substitute $p = \frac{\lambda}{n}$ into the Binomial Probability Mass Function and take the limit as $n \to \infty$:

$$\lim_{n\to\infty} P(Y = k) = \lim_{n\to\infty} \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \lim_{n\to\infty} \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

We can group the constants (with respect to n) and pull them to the front:

$$= \frac{\lambda^k}{k!} \lim_{n\to\infty} \left[\frac{n!}{(n-k)!} \cdot \frac{1}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k} \right]$$

Deriving the Poisson PMF: Evaluation

Let's break the expression inside the limit into three distinct components:

$$= \frac{\lambda^k}{k!} \lim_{n\to\infty} \underbrace{\left[\frac{n(n-1)\dots(n-k+1)}{n^k}\right]}_{\text{Part A}} \underbrace{\left(1-\frac{\lambda}{n}\right)^n}_{\text{Part B}} \underbrace{\left(1-\frac{\lambda}{n}\right)^{-k}}_{\text{Part C}}$$

1. **Part A:** There are exactly k terms in the numerator polynomial and n^k in the denominator. As $n \to \infty$, the highest degree terms dominate, and this fraction approaches 1.
2. **Part B:** From standard calculus, the limit definition of the exponential function dictates that $\lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n = e^{-\lambda}$.
3. **Part C:** For any finite k, as $n \to \infty$, the term inside the parentheses approaches 1. Thus, $1^{-k} = 1$.

The Poisson Distribution

Multiplying our evaluated limits together $(1 \cdot e^{-\lambda} \cdot 1)$, we arrive at the PMF for a new, fundamental distribution.

The Poisson PMF

A random variable X follows a Poisson distribution with rate parameter $\lambda > 0$, denoted $X \sim \text{Poisson}(\lambda)$, if:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \text{for } k = 0, 1, 2, \ldots$$

Mean and Variance of the Poisson: Intuitive

Because we derived this by holding $np = \lambda$ constant:

- **Mean:** $E[X] = \lim_{n\to\infty}(np) = \lambda$
- **Variance:** $Var(X) = \lim_{n\to\infty} np(1-p) = \lambda(1-0) = \lambda$

Poisson Expectation: Direct Derivation

Let $X \sim \text{Poisson}(\lambda)$. By definition, the expected value is:

$$E[X] = \sum_{k=0}^{\infty} kP(X = k) = \sum_{k=0}^{\infty} k\frac{\lambda^k e^{-\lambda}}{k!}$$

The term for $k = 0$ is exactly zero, so we can start the sum at $k = 1$. We cancel k with $k!$, leaving $(k-1)!$ in the denominator:

$$E[X] = \sum_{k=1}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-1)!}$$

We factor out $\lambda e^{-\lambda}$ to shift the index. Let $j = k - 1$:

$$E[X] = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Recognizing the series expansion for $e^{\lambda} = \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$:

$$E[X] = \lambda e^{-\lambda}(e^{\lambda}) = \lambda \quad \blacksquare$$

Poisson Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[X(X-1)]$:

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\frac{\lambda^k e^{-\lambda}}{k!}$$

The terms for $k = 0$ and $k = 1$ are zero. Starting the sum at $k = 2$, we cancel $k(k-1)$ with the $k!$ in the denominator:

$$E[X(X-1)] = \sum_{k=2}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-2)!}$$

Factor out $\lambda^2 e^{-\lambda}$ and let $j = k - 2$:

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Once again, the infinite sum resolves exactly to e^{λ}:

$$E[X(X-1)] = \lambda^2 e^{-\lambda}(e^{\lambda}) = \lambda^2$$

Poisson Variance: Direct Derivation

We now have the two necessary components: $E[X(X-1)] = \lambda^2$ and $E[X] = \lambda$.

1. Find the Second Moment $E[X^2]$

Expand the factorial moment: $E[X(X-1)] = E[X^2] - E[X]$.

Rearranging gives us:

$$\begin{aligned} E[X^2] &= E[X(X-1)] + E[X] \\ &= \lambda^2 + \lambda \end{aligned}$$

2. Calculate Final Variance

Apply the standard variance formula $Var(X) = E[X^2] - (E[X])^2$:

$$\begin{aligned} Var(X) &= (\lambda^2 + \lambda) - (\lambda)^2 \\ &= \lambda^2 + \lambda - \lambda^2 \\ &= \lambda \quad \blacksquare \end{aligned}$$

Waiting Time: The Geometric Distribution

We return to our sequence of independent Bernoulli trials (where $P(\text{Success}) = p$ and $P(\text{Failure}) = q = 1 - p$).
Instead of counting the *number* of successes in a fixed time, we now ask:
How long must we wait until the very first success?

Defining the Random Variable

Let X be the number of trials needed to get the first success.

- The only way for $X = k$ to occur is to have exactly $k - 1$ failures followed immediately by 1 success.
- Because the trials are independent, we multiply their probabilities.

The Geometric PMF

We say $X \sim \text{Geometric}(p)$. Its Probability Mass Function is:

$$P(X = k) = q^{k-1}p, \quad \text{for } k = 1, 2, 3, \ldots$$

Geometric Expectation:

By definition, $E[X] = \sum_{k=1}^{\infty} kP(X = k) = \sum_{k=1}^{\infty} kq^{k-1}p$.
We can pull the constant p outside the sum:

$$E[X] = p\sum_{k=1}^{\infty} kq^{k-1}$$

To evaluate this infinite series, we recall the standard geometric series for $|q| < 1$:

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$$

If we take the derivative of both sides with respect to q, the k drops down:

$$\frac{d}{dq}\sum_{k=0}^{\infty} q^k = \sum_{k=1}^{\infty} kq^{k-1} = \frac{d}{dq}(1-q)^{-1} = \frac{1}{(1-q)^2}$$

Substituting this result back into our expectation formula:

$$E[X] = p\left[\frac{1}{(1-q)^2}\right] = p\left[\frac{1}{p^2}\right] = \frac{1}{p} \quad \blacksquare$$

Geometric Variance and Intuition

To find the variance, we would use the second derivative of the geometric series to find the second factorial moment $E[X(X-1)]$.

Moments Summary

For $X \sim \text{Geometric}(p)$:

- **Mean:** $E[X] = \frac{1}{p}$
- **Variance:** $Var(X) = \frac{1-p}{p^2}$

Physical Intuition

The expected value $E[X] = 1/p$ aligns perfectly with human intuition:

- If you flip a fair coin ($p = 1/2$), you expect to wait $1/(1/2) = 2$ flips to see the first Heads.
- If you roll a fair die looking for a six ($p = 1/6$), you expect to wait $1/(1/6) = 6$ rolls to see the first 6.

The Memoryless Property

A unique and defining characteristic of the Geometric distribution is that it is **memoryless**.

Formal Definition

For any integers $s > 0$ and $t > 0$:

$$P(X > s + t \mid X > s) = P(X > t)$$

The probability of waiting t additional trials for a success, given that we have already waited s trials, is exactly the same as the probability of waiting t trials from the very beginning.

The samples do not "remember" past failures. Suppose you are rolling a fair die waiting for a 6.

- You have already rolled 10 times and failed.
- The die is not "due" for a 6. The probability that it takes 3 *more* rolls is exactly the same as taking 3 rolls on a brand new die.

Proof of the Memoryless Property

Prerequisite: Check that $P(X > k) = q^k$. (To fail to get a success in the first k trials means getting k failures in a row).

Using the definition of conditional probability:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t \text{ and } X > s)}{P(X > s)}$$

Since the event $\{X > s + t\}$ is a subset of the event $\{X > s\}$, their intersection is simply $\{X > s + t\}$. We substitute our tail probabilities:

$$\begin{aligned} P(X > s + t \mid X > s) &= \frac{P(X > s + t)}{P(X > s)} \\ &= \frac{q^{s+t}}{q^s} \\ &= q^t \\ &= P(X > t) \quad \blacksquare \end{aligned}$$

Conclusion: The past history of failures algebraically cancels out.

Exercise: Which Distribution?

Let's test our intuition. For each scenario, identify the most appropriate distribution model and its parameters.

1. The Single Seed

A plant seed has a 25% chance to sprout with green leaves. We plant exactly one seed. Let $X = 1$ if it is green, and $X = 0$ if it is not.

2. The Sales Calls

A worker makes 50 phone calls. Each call has an independent 5% chance of resulting in a sale. Let X be the total number of sales made today.

3. The Random Inspection

A computer randomly picks one shipping box out of 12 to inspect. Every box is equally likely to be picked. Let X be the chosen box number.

Exercise: Which Distribution

Identify the most appropriate distribution model and its parameters.

4. Summer Storms

Historical weather data shows that a specific town gets hit by lightning an average of 14 times per summer. Let X be the total number of lightning strikes this coming summer.

5. The Rare Blood Type

Only 2% of blood donors have a specific rare antibody. A doctor tests donated blood bags one by one. Let X be the total number of bags tested until they find the **very first** match.

Exercise

1. The Environmental Sensor

An oceanic sensor is designed to detect a specific microplastic particle. When a particle passes the sensor, there is a 10% chance the sensor successfully registers it.

- How many particles must pass by the sensor to ensure the probability of detecting **at least one** particle is strictly greater than 95%?

2. The Bakery Orders

A bakery receives an average of 2.5 large catering orders per week. Assume the weekly number of catering orders follows a Poisson distribution.

- **(a)** What is the probability the bakery receives at least 2 catering orders this week?
- **(b)** If we know the bakery received *at least one* order this week, what is the exact probability it received exactly 3 orders?

Exercise

3. The Tabletop Game

In a role-playing game, a player rolls a fair 20-sided die. They need to roll a "Natural 20" (a probability of 1/20) to break a curse. They keep rolling until they succeed.

- What is the probability it takes **more than** 5 rolls to break the curse?

4. The Mystery Raffle

A charity raffle sells consecutively numbered tickets starting from Ticket #1 up to Ticket #N. The winning ticket is drawn with perfect fairness. If the **expected value** of the winning ticket number is 42:

- **(a)** How many total tickets (N) were sold?
- **(b)** What is the variance of the ticket numbers?

From Discrete to Continuous

Until now, our random variables jumped between distinct, separate values (counts, dice rolls, iterations). Now, we enter the continuous realm, where a variable can take **any real number** within a range (like exact weight, continuous time, or distance).

The Probability Density Function (PDF)

Instead of a Probability Mass Function (PMF), continuous variables are described by a PDF, denoted as $f(x)$.

- $f(x)$ is **not** a probability itself. It is a density.
- Probabilities are calculated by finding the **area under the curve** over a specific interval.

Because a continuous range has infinitely many points, the probability of hitting one exact, infinitely precise number is always zero:

$$P(X = c) = \int_c^c f(x)dx = 0$$

Deriving the Exponential: The Setup

Assume we have a continuous random variable $X \geq 0$ that possesses the memoryless property:

$$P(X > s + t \mid X > s) = P(X > t) \quad \text{for all } s, t \geq 0$$

Using the definition of conditional probability:

$$\frac{P(X > s + t \text{ and } X > s)}{P(X > s)} = P(X > t)$$

Since $(X > s + t)$ is a strict subset of $(X > s)$, their intersection is just $(X > s + t)$:

$$\frac{P(X > s + t)}{P(X > s)} = P(X > t) \implies P(X > s + t) = P(X > s)P(X > t)$$

To make this easier to read, let us define the **tail probability** (or survival function) as $G(x) = P(X > x)$. We now have a strict functional equation:

$$G(s + t) = G(s)G(t)$$

Deriving the Exponential: The Solution

We need to find a tail probability function $G(x)$ that satisfies our memoryless equation:

$$G(s+t) = G(s)G(t)$$

Finding the Function

The standard algebraic function that converts addition in its input to multiplication in its output is the exponential function:

$$e^{\lambda(s+t)} = e^{\lambda s} \cdot e^{\lambda t}$$

Because $G(x) = P(X > x)$ represents a probability, it has strict boundaries:

- It must start at $G(0) = 1$.
- It must strictly decrease as the waiting time x increases.

To make the function decay rather than grow infinitely, the exponent must be negative. Thus, for some rate parameter $\lambda > 0$:

$$G(x) = e^{-\lambda x}$$

The Exponential Distribution

We just proved that the tail probability must be $P(X > x) = e^{-\lambda x}$.

1. Find the CDF

The Cumulative Distribution Function, $F(x) = P(X \leq x)$, is simply the complement of the tail probability:

$$F(x) = 1 - P(X > x) = 1 - e^{-\lambda x}$$

2. Find the PDF

By the Fundamental Theorem of Calculus, the Probability Density Function $f(x)$ is the derivative of the CDF:

$$\begin{aligned} f(x) &= \frac{d}{dx} F(x) \\ &= \frac{d}{dx} \left(1 - e^{-\lambda x}\right) \\ &= -e^{-\lambda x} \cdot (-\lambda) \\ &= \lambda e^{-\lambda x} \quad \blacksquare \end{aligned}$$

Exponential Distribution: Mean Derivation

Let $X \sim \text{Exponential}(\lambda)$ with PDF $f(x) = \lambda e^{-\lambda x}$. By definition:

$$E[X] = \int_0^\infty x\lambda e^{-\lambda x} dx$$

We use Integration by Parts (IBP): $\int u\, dv = uv - \int v\, du$.

- Let $u = x \implies du = dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$\begin{aligned} E[X] &= \left[-xe^{-\lambda x}\right]_0^\infty - \int_0^\infty (-e^{-\lambda x}) dx \\ &= (0 - 0) + \int_0^\infty e^{-\lambda x} dx \quad \left(\text{Note: } \lim_{x\to\infty} \frac{x}{e^{\lambda x}} = 0 \text{ by L'Hôpital's}\right) \\ &= \left[-\frac{1}{\lambda} e^{-\lambda x}\right]_0^\infty \\ &= 0 - \left(-\frac{1}{\lambda}\right) = \frac{1}{\lambda} \quad \blacksquare \end{aligned}$$

Exponential Distribution: Second Moment

To find the variance, we first need $E[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x} dx$. Using IBP again:

- Let $u = x^2 \implies du = 2x\,dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$E[X^2] = \left[-x^2 e^{-\lambda x}\right]_0^\infty - \int_0^\infty (-e^{-\lambda x}) \cdot 2x\,dx$$
$$= 0 + 2\int_0^\infty x e^{-\lambda x} dx$$

Notice that the remaining integral is almost $E[X]$. We multiply and divide by λ:

$$E[X^2] = \frac{2}{\lambda} \int_0^\infty x \lambda e^{-\lambda x} dx$$
$$= \frac{2}{\lambda} \cdot E[X] = \frac{2}{\lambda} \cdot \frac{1}{\lambda} = \frac{2}{\lambda^2}$$

Exponential Distribution: Variance

Now we apply the variance formula:

$$Var(X) = E[X^2] - (E[X])^2$$

The Final Calculation

Substituting our derived moments:

$$\begin{aligned} Var(X) &= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 \\ &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} \\ &= \frac{1}{\lambda^2} \quad \blacksquare \end{aligned}$$

Summary for $X \sim \text{Exp}(\lambda)$

- **Mean:** $E[X] = 1/\lambda$
- **Standard Deviation:** $SD(X) = 1/\lambda$